

CBSE Class 12
Chemistry (Theory)
Previous Year Question Paper 2016

Series: ONS

Set -1

Code no. 56/1/C

- Please check that this question paper contains **16** printed pages.
- Code number given on the right-hand side of the question paper should be written on the title page of the answer-book by the candidate.
- Please check that this question paper contains **26** questions.
- **Please write down the Serial Number of the question before attempting it.**
- 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m, the students will read the question paper only and will not write any answer on the answer book during this period

Chemistry (Theory)

Time Allowed: 3 hours

Maximum Marks: 70

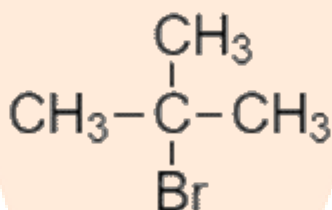
General Instructions:

1. **All** questions are compulsory.
2. Question number **1 to 5** are very short-answer questions and carry **1** mark each.
3. Question number **6 to 10** are short-answer questions and carry **2** marks each.
4. Question number **11 to 22** are also short-answer questions and carry **3** marks each.

5. Question number **23** is a value based question and carries **4** marks.
6. Question number **24 to 26** are long-answer questions and carry **5** marks each.
7. Use log tables, if necessary. Use of calculators is **not** allowed.

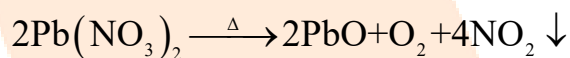
1. Write the structure of an isomer of compound C_4H_9Br which is most reactive towards S_N1 reaction **1 Mark**

Ans:

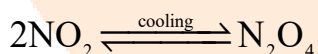


2. $Pb(NO_3)_2$ on heating gives a brown gas which undergoes dimerization on cooling ? Identify the gas. **1 Mark**

Ans: The gas is NO_2 (Nitrogen dioxide).



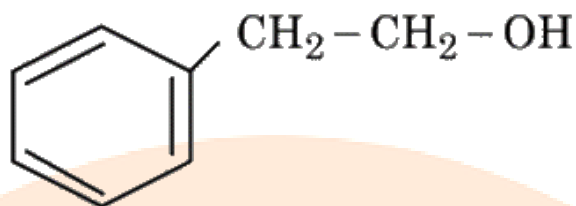
Brown gas



3. Give an example of a molecular solid and an ionic solid. **1 Mark**

Ans: The example of a molecular solid is I_2 and the example of an ionic solid is $NaCl$.

4. Write the IUPAC name of the given compound. **1 Mark**



Ans: The IUPAC name of the given compound is 2-Phenylethanol.

5. What is the reason for the stability of colloidal sols?

1 Mark

Ans: The charge of all colloidal particles in a solution is the same, and the charge of the dispersion medium is the opposite and equal, making the system electrically neutral. This is the primary cause for the colloidal solution's stability.

6. (i) Gas (A) is more soluble in water than Gas (B) at the same temperature. Which one of the two gases will have the higher value of K_H (Henry's constant) and why ?

2 Marks

Ans: Because the solubility of the gas in the liquid is smaller, the value of K_H (Henry's constant) will be larger for Gas B.

(ii) In a non-ideal solution, what type of deviation shows the formation of maximum boiling azeotropes?

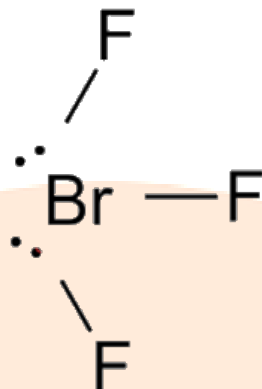
Ans: Negative deviation indicates the production of maximum boiling azeotropes in non-ideal solutions.

7. Write the structure of the following:

2 Marks

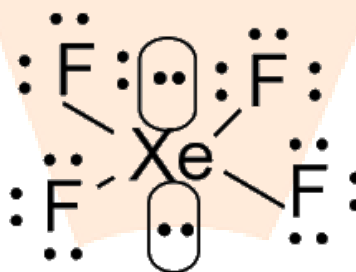
(i) BrF_3

Ans: The structure is:



(ii) XeF_4

Ans:

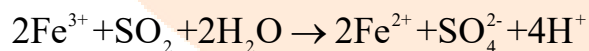


Or

What happens when:

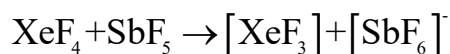
(i) SO_2 gas is passed through an aqueous solution Fe^{3+} salt.

Ans: When SO_2 gas is passed through an aqueous solution of Fe^{3+} salt, SO_2 acts as a reducing agent, reducing Fe^{3+} to Fe^{2+} and turning the brown colour of the iron (III) solution green.



(ii) XeF_4 reacts with SbF_5 ?

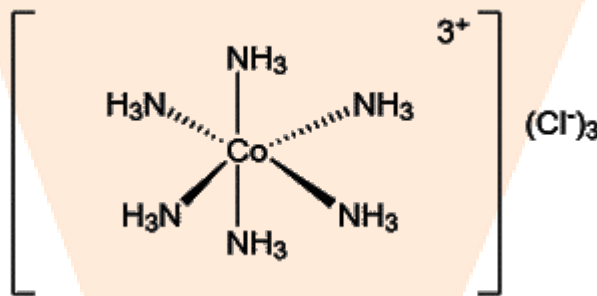
Ans: Complexes (addition compounds) are formed when XeF_2 reacts with strong Lewis acids such as SbF_5 .



8. When a coordination compound $\text{CoCl}_3 \cdot 6\text{NH}_3$ is mixed with AgNO_3 , 3 moles of AgCl are precipitated per mole of the compound. Write 2 Marks

(i) Structural formula of the complex

Ans: $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$



(ii) IUPAC name of the complex

Ans: Hexaamminecobalt(III)chloride

9. For a reaction: $\text{H}_2 + \text{Cl}_2 \xrightarrow{h\nu} 2\text{HCl}$ Rate = K 2 Marks

(i) Write the order and molecularity of this reaction.

Ans: The order of this reaction is zero-order and its molecularity is 2.

(ii) Write the unit of K.

Ans: $\text{mol L}^{-1} \text{s}^{-1}$

10. Write the chemical equations involved in the following reactions : 2 Marks

(i) Hoffmann-bromamide degradation reaction

Ans: $\text{Ar/R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2 + \text{Br}_2 + 4\text{NaOH} \rightarrow \text{Ar/R}-\text{NH}_2 + \text{Na}_2\text{CO}_3 + 2\text{NaBr} + 2\text{H}_2\text{O}$

Here R = alkyl group and Ar = aryl group

(ii) Carbylamine reaction



Here R = alkyl group and Ar = aryl group

11. An element crystallizes in a b.c.c. lattice with cell edge of 500pm. The density of the element is 7.5g cm^{-3} . How many atoms are present in 300g of the element?
3 Marks

Ans: $z=2$

$$d = \frac{z \times M}{a^3}$$

$$7.5 = \frac{2 \times M}{(500)^3 \times 10^{-30} \times 6.022 \times 10^{23}}$$

$$M = \frac{7.5 \times 125 \times 10^{-24} \times 6.022 \times 10^{23}}{2}$$

$$= 282.3 \text{ g mol}^{-1}$$

$$282.3 \text{ g} = 6.022 \times 10^{23} \text{ atoms}$$

$$300 \text{ g} = \frac{6.022 \times 10^{23} \times 300}{282.3}$$

$$= 6.022 \times 10^{23} \text{ atoms}$$

12. For the first order thermal decomposition reaction, the following data were obtained :
3 Marks



Time / sec	Total pressure / atm
0	0.30
300	0.50

Calculate the rate constant

(Given : $\log 2 = 0.301$, $\log 3 = 0.4771$, $\log 4 = 0.6201$)

Ans: Given: Initial pressure

$$P_0 = 0.30 \text{ atm}, P_t = 0.50 \text{ atm}, t = 300 \text{ s}$$

$$\text{Rate constant, } k = \frac{2.303}{t} \log \frac{P_0}{2P_0 - P_t}$$

$$= \frac{2.303}{300 \text{ s}} \log \frac{0.30}{2 \times 0.30 - 0.50}$$

$$= \frac{2.303}{300 \text{ s}} \log \frac{0.30}{0.60 - 0.50}$$

$$= \frac{2.303}{300 \text{ s}} \log \frac{0.30}{0.10}$$

$$= \frac{2.303}{300 \text{ s}} \log 3$$

$$= \frac{2.303}{300 \text{ s}} \times 0.4771$$

$$= \frac{1.099}{300 \text{ s}}$$

$$= 0.0036 \text{ s}^{-1} \text{ or } 3.6 \times 10^{-3} \text{ s}^{-1}$$

13. Define the following terms:

3 Marks

(i) Lyophilic colloid

Ans: Lyophilic sols are colloidal sols formed directly by combining substances like gum, gelatine, starch, rubber, and others with a suitable liquid (the dispersion medium), such as muddy water. They are quite stable and cannot be coagulated easily.

(ii) Zeta potential

Ans: Helmholtz electrical double layer refers to the combination of two layers of opposite charges surrounding colloidal particles. The first layer of ions is firmly held and is referred to as the fixed layer, whereas the second layer is mobile and is referred to as the diffused layer. Because charge separation is a site of potential difference between opposite charges, it is also known as zeta potential or electrokinetic potential.

(iii) Associated colloids

Ans: Some substances behave as normal strong electrolytes at low concentrations but exhibit colloidal behaviour at higher concentrations due to aggregate formation. The particles that form as a result of this process are known as associated colloids or micelles.

14.

3 Marks

(i) Name the method of refining of nickel

Ans: Nickel is refined by using Mond's process.

(ii) What is the role of cryolite in the extraction of aluminium?

Ans: The melting point of alumina is very high. It is dissolved in the cryolite which lowers the melting point and brings conductivity.

(iii) What is the role of limestone in the extraction of iron from its oxides?

Ans: The role of limestone in the extraction of iron from its oxides is to remove silica impurity of the ore as a slag.

15. Calculate the boiling point of the solution when 4 g of MgSO_4 ($M = 120 \text{ g mol}^{-1}$) was dissolved in 100 g of water, assuming MgSO_4 undergoes complete ionization. (K_b for water = $0.52 \text{ K kg mol}^{-1}$) **3 Marks**

Ans: $\Delta T_b = iK_b m$

$i=1$

$$= i \times K_b \times \frac{W_2 \times 1000}{M \times W_1}$$

$$= 2 \times 0.52 \text{ K kg mol}^{-1} \times \frac{4 \text{ g} \times 1000 \text{ g kg}^{-1}}{120 \text{ g mol}^{-1} \times 100 \text{ g}}$$

$$= \frac{2 \times 0.52}{3}$$

$$= 0.346 \text{ K}$$

$$\text{Boiling point of water} = \frac{373.15 \text{ K}}{737 \text{ K}}$$

$$T_b = T_b^\circ + \Delta T_b$$

$$= 373.15 \text{ K} + 0.346 \text{ K} \text{ or } 737 \text{ K} + 0.346 \text{ K}$$

$$= 373.496 \text{ K} \text{ or } 373.346 \text{ K}$$

16. Give reasons :

3 Marks

(i) SO_2 is reducing while TeO_2 is an oxidizing agent.

Ans: SO_2 is reducing while TeO_2 is an oxidizing agent because stability of higher oxidation state decreases as we move down the group. S is more stable in (+6) oxidation state whereas Te is more stable in (+4) oxidation state.

(ii) Nitrogen does not form pentahalide.

Ans: Nitrogen does not form pentahalide due to the absence of d orbital.

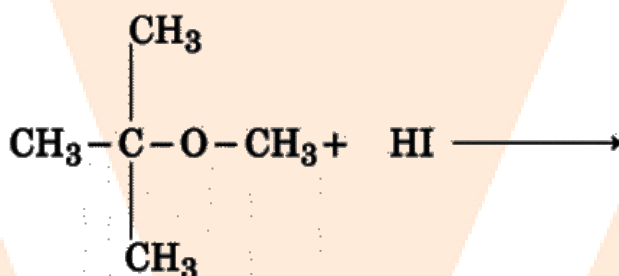
(iii) ICl is more reactive than I₂.

Ans: Compounds containing interhalogens ICl are more reactive than I₂ because the CI bond in inter halogens is weaker than the bonds in halogens, with the exception of F-F bonds.

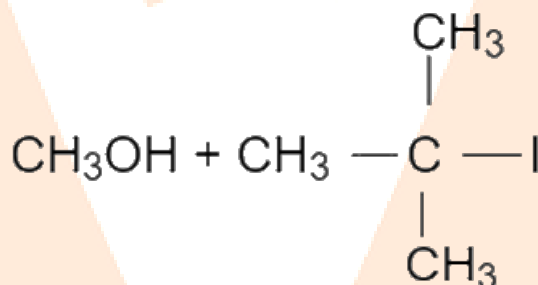
17. Write the final product (s) in each of the following reactions:

3 Marks

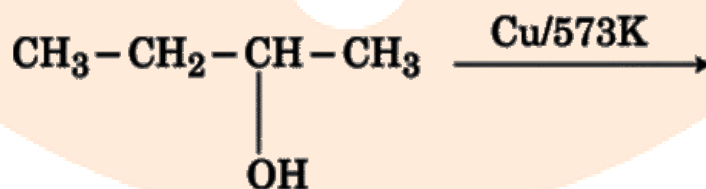
(a)



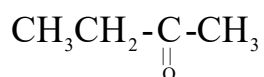
Ans: The final product is:

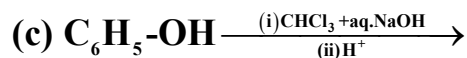


(b)

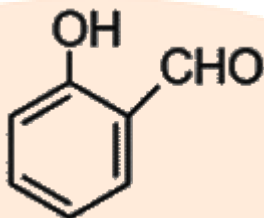


Ans: The final product is:





Ans: The final product is:

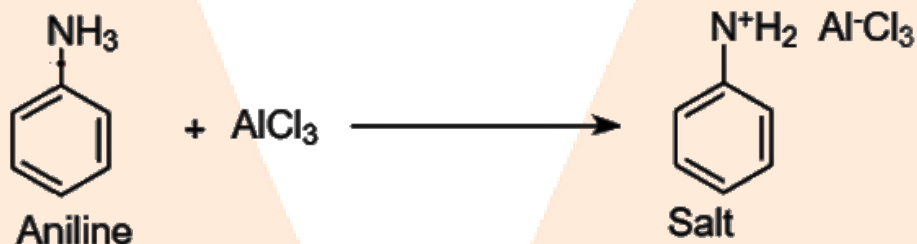


18. Give reasons for the following:

3 Marks

(i) Aniline does not undergo Friedel-Crafts reaction.

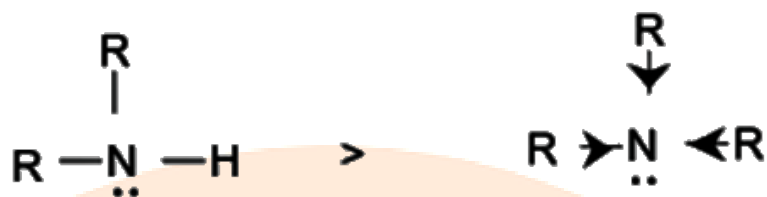
Ans: In the presence of AlCl_3 , a Friedel Crafts reaction is carried out. However, AlCl_3 , which is used as a catalyst, is acidic in nature, whereas aniline is a strong base. As a result, aniline reacts with AlCl_3 to form a salt.



Electrophilic substitution in the benzene ring is deactivated due to the positive charge on the N—atom. As a result, aniline does not undergo the Friedel—Crafts reaction.

(ii) $(\text{CH}_3)_2\text{NH}$ is more basic than $(\text{CH}_3)_3\text{N}$ in an aqueous solution.

Ans: In an aqueous solution, $(\text{CH}_3)_2\text{NH}$ is more basic than $(\text{CH}_3)_3\text{N}$. The +I effect will increase in the alkyl group, increasing the likelihood of lone pair electron donation. Amine accepts a proton and a cation, both of which are stabilised in water by solvation. The greater the solvation by hydrogen bonding, the greater the basic strength.



As a result, as the methyl group increases, so does hydrogen bonding and solvation stabilisation. This has the net effect of decreasing basic strength from secondary to tertiary amine.

(iii) Primary amines have higher boiling point than tertiary amines.

Ans: Tertiary amines contain no H-atoms, whereas primary amines contain two H-atoms. Primary amines exhibit extensive intermolecular H-bonding due to the presence of H-atoms.



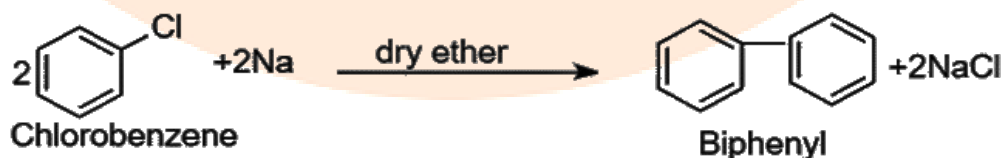
As a result, more energy is required to separate the primary amine molecules. As a result, primary amines have a higher boiling point than secondary amines.

19. How do you convert:

3 Marks

(i) Chlorobenzene to biphenyl

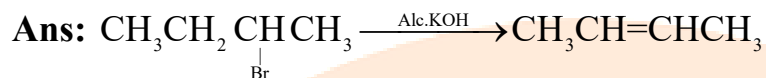
Ans:



(ii) Propene to 1-iodopropane



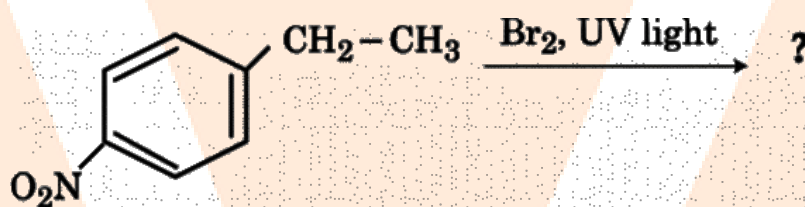
(iii) 2-bromobutane to but-2-ene



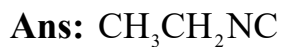
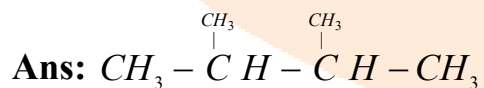
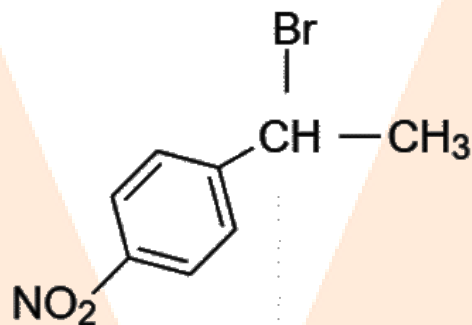
Or

Write the major product(s) in the following :

(i)



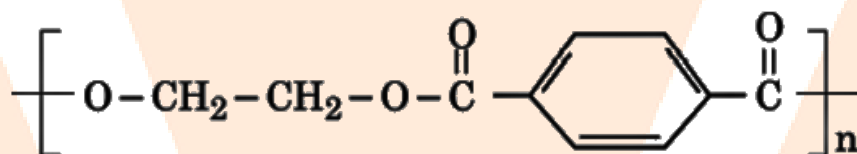
Ans:



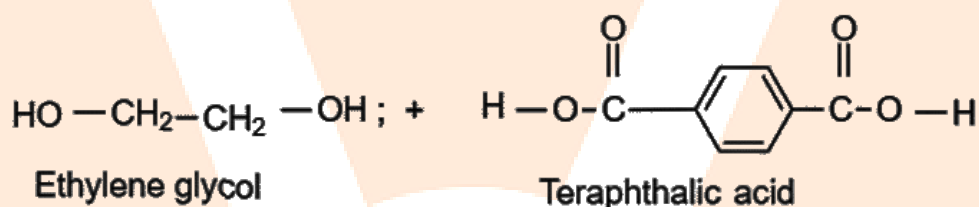
20.(i) What is the role of Sulphur in the vulcanization of rubber? 3 Marks

Ans: Rubber's physical properties can be modified to meet specific needs by incorporating sulphur into the vulcanization process. The process of introducing -S-S- polysulphide crosslinks between adjacent chains tends to limit chain motion relative to one another.

(ii) Identify the monomers in the following polymer:



Ans: The monomers in the following polymers are:



(iii) Arrange the following polymers in the increasing order of their intermolecular forces :

Terylene, Polythene, Neoprene

Ans: Neoprene < Polythene < Terylene

21. (i) Write the structural difference between starch and cellulose. 3 Marks

Ans: The structural difference between starch and cellulose is given below:

Starch	Cellulose
The glucose monomers in starch are in an alpha configuration.	The glucose monomers in cellulose are in a beta configuration.
Starch is a polymer made up of	Cellulose is a long chain made up

amylose and amylopectin.

entirely of β -D-glucose units.

(ii) What type of linkage is present in Nucleic acids?

Ans: Phosphodiester linkage in which the 5'-phosphate group of one nucleotide unit is joined to the 3'-hydroxyl group of the next nucleotide.

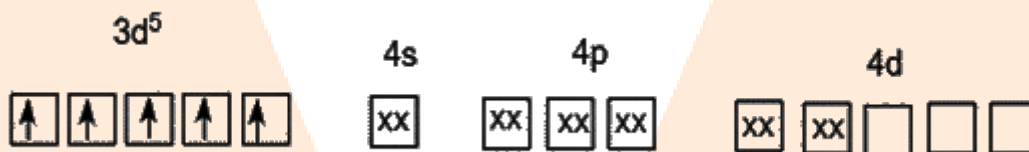
(iii) Give one example each for fibrous protein and globular protein.

Ans: The example of fibrous protein is silk.

The example of globular protein is egg albumin.

22. (a) For the complex $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, write the hybridization, magnetic character and spin of the complex. (At. number : Fe=26) **3 Marks**

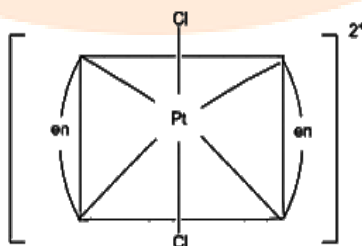
Ans: $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ has sp^3d^2 hybridization.



The oxidation state of Fe is +3 (i.e. in d^5 configuration). Because water is a weak ligand, there is no paired electron. As a result, it is strongly paramagnetic and has a high spin (octahedral geometry).

(b) Draw one of the geometrical isomers of the complex $[\text{Pt}(\text{en})_2\text{Cl}_2]^{2+}$ which is optically inactive.

Ans:



23. Due to hectic and busy schedule, Mr. Singh started taking junk food in the lunch break and slowly became habitual of eating food irregularly to excel in his field. One day during meeting he felt severe chest pain and fell down. Mr. Khanna, a close friend of Mr. Singh, took him to doctor immediately. The doctor diagnosed that Mr. Singh was suffering from acidity and prescribed some medicines. Mr. Khanna advised him to eat home made food and change his lifestyle by doing yoga, meditation and some physical exercise. Mr. Singh followed his friend's advice and after a few days he started feeling better.

4 Marks

After reading the above passage, answer the following:

(i) What are the values (at least two) displayed by Mr. Khanna?

Ans: Caring nature and being supportive are the two values displayed by Mr. Khanna.

(ii) What are antacids? Give one example.

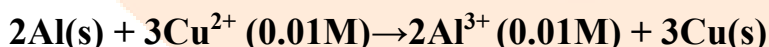
Ans: Antacids are medications that are used to reduce stomach acidity. For example- mixture of aluminum and magnesium hydroxide.

(iii) Would it be advisable to take antacids for a long period of time ? Give reason

Ans: No, it is not advisable to take antacids for a long period of time because excessive antacid can make the stomach alkaline and trigger the production of more acid.

24. (a) Calculate E° for the following reaction at 298 K :

5 Marks



Given : $E_{\text{cell}} = 1.98\text{V}$

Ans:

$$E_{\text{cell}}^\circ = 1.98\text{V} + \frac{0.591}{6} \log$$

$$E_{\text{cell}}^{\circ} = E_{\text{cell}}^{\circ} + \frac{0.591}{n} \log \frac{[\text{Al}^{3+}]^2}{[\text{Cu}^{2+}]^3}$$

$$E_{\text{cell}}^{\circ} = 1.98\text{V} + \frac{0.591}{6} \log \frac{(0.01)^2}{(0.01)^3}$$

$$E_{\text{cell}}^{\circ} = 1.98\text{V} + \frac{0.591}{6} \log 10^2$$

$$E_{\text{cell}}^{\circ} = 1.98\text{V} + \frac{0.591}{6} \times 2 \times \log 10$$

$$E_{\text{cell}}^{\circ} = 1.98\text{V} + \frac{0.591}{6} \times 2$$

$$E_{\text{cell}}^{\circ} = 1.98\text{V} + 0.0197\text{V}$$

$$E_{\text{cell}}^{\circ} = 1.9997\text{V}$$

(b) Using the E° values of A and B, predict which is better for coating the surface of iron [$E^{\circ}(\text{Fe}^{2+}/\text{Fe}) = -0.44\text{V}$] to prevent corrosion and why?

Given: $E^{\circ}(\text{A}^{2+}/\text{A}) = -2.37\text{V}$; $E^{\circ}(\text{B}^{2+}/\text{B}) = -0.14\text{V}$

Ans: A is better for coating the surface of iron to prevent corrosion because its E° value is more negative.

Or

(a) The conductivity of 0.001 mol L^{-1} solution of CH_3COOH is $3.905 \times 10^{-5}\text{ S cm}^{-1}$. Calculate its molar conductivity and degree of dissociation (α)

Given $\lambda^{\circ}(\text{H}^{+}) = 349.6\text{ S cm}^2\text{ mol}^{-1}$ and $\lambda^{\circ}(\text{CH}_3\text{COO}^{-}) = 40.9\text{ S cm}^2\text{ mol}^{-1}$

Ans:

$$C = 0.001\text{ mol L}^{-1}, K = 3.905 \times 10^{-5}\text{ S cm}^{-1}$$

Molar Conductivity

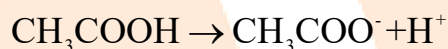
$$\lambda_m = \frac{k \times 1000}{c}$$

$$= \frac{3.905 \times 10^{-5} \times 1000}{0.001}$$

$$= 39.05 \text{ S cm}^2 \text{ mol}^{-1} \text{ or } \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

$$\lambda^\circ(\text{H}^+) = 349.6 \text{ S cm}^2 \text{ mol}^{-1}$$

$$\lambda^\circ(\text{CH}_3\text{COO}^-) = 40.9 \text{ S cm}^2 \text{ mol}^{-1}$$



$$\lambda^\circ_{(\text{CH}_3\text{COO}^-)} = \lambda^\circ \text{CH}_3\text{COO}^- + \lambda^\circ \text{H}^+$$

$$= (40.9 + 349.6) \text{ S cm}^2 \text{ mol}^{-1}$$

$$= 390.5 \text{ S cm}^2 \text{ mol}^{-1}$$

$$\alpha = \frac{\lambda_m}{\lambda^\circ_m} = \frac{39.05}{390.5}$$

$$= 0.1$$

(b) Define electrochemical cell. What happens if external potential applied becomes greater than E°_{cell} of an electrochemical cell ?

Ans: An electrochemical cell is a device that generates electricity from the energy emitted during a spontaneous chemical reaction and utilizes it to effect a chemical change.

When the external potential applied to an electrochemical cell is larger than the E°_{cell} , the reaction is reversed, and the cell becomes an electrolytic cell, and vice versa.

25.

5 Marks

(a) Account for the following :

(i) Mn shows the highest oxidation state of +7 with oxygen but with fluorine it shows the highest oxidation state of +4

Ans: Manganese can form a $p\pi - d\pi$ bond with oxygen by using the 2p-orbital of oxygen and the 3d-orbital of manganese, resulting in the highest oxidation state of +7. Because fluorine cannot form such a $p\pi-d\pi$ bond, it can only have a maximum oxidation state of +4.

(ii) Cr^{2+} is a strong reducing agent.

Ans: Cr^{2+} has a stable d^4 configuration, whereas Cr^{3+} has a less stable d^5 configuration. As a result of the greater stability of the +3 oxidation state, Cr has a proclivity to acquire Cr^{3+} . Cr^{2+} is a reducing agent.

(iii) Cu^{2+} salts are coloured while Zn^{2+} salts are white.

Ans: The d-subshell has been completely filled by the Zn^{2+} ion, and no d-d transition is possible. As a result, zinc salts are white. Cu^{2+} has the configuration $[\text{Ar}] 3d^9$. It has a partially filled d-subshell and thus is coloured due to the d-d transition.

(b) Complete the following equations:

(i) $2\text{MnO}_2 + 4\text{KOH} + \text{O}_2 \xrightarrow{\Delta}$

Ans: $2\text{MnO}_2 + 4\text{KOH} + \text{O}_2 \xrightarrow{\Delta} 2\text{K}_2\text{MnO}_4 + 2\text{H}_2\text{O}$

(ii) $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{I}^- \rightarrow$

Ans: $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{I}^- \rightarrow 2\text{Cr}^{3+} + 3\text{I}_2 + 7\text{H}_2\text{O}$

Or

The elements of 3 d transition series are given as :

Sc Ti V Cr Mn Fe Co Ni Cu Zn

Answer the following:

(i) Write the element which shows the maximum number of oxidation states. Give a reason.

Ans: Manganese (Mn) has the most oxidation states of any element

Element	Possible Oxidation State	More Stable Oxidation State
Manganese	+2, +3, +4, +5, +6, +7	+2, +7

Mn has the electronic configuration $[\text{Ar}]4s^23d^5$. As a result, there are 7 electrons that can be shared.

(ii) Which element has the highest m.p?

Ans: Because Cr has the configuration $[\text{Ar}]3d^54s^1$, 6 unpaired electrons participate in the formation of metallic bonds, it has the highest melting point.

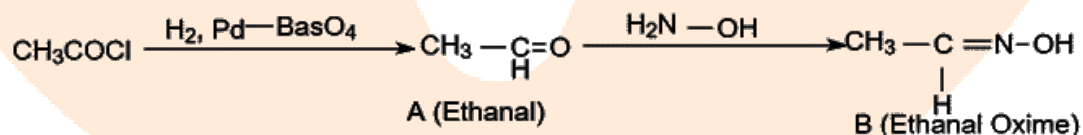
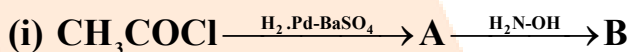
(iii) Which element shows only +3 oxidation state ?

Ans: Scandium(Sc) shows only +3 oxidation state.

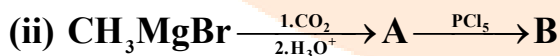
(iv) Which element is a strong oxidizing agent in +3 oxidation state and why?

Ans: Mn is a strong oxidizing agent in +3 oxidation state because the transition from Mn^{3+} to Mn^{2+} results in a stable half filled (d^5) electronic configuration.

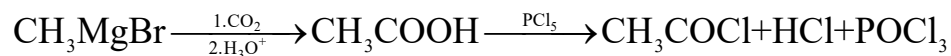
26. (a) Write the structures of A and B in the following reactions:



Ans:



Ans:



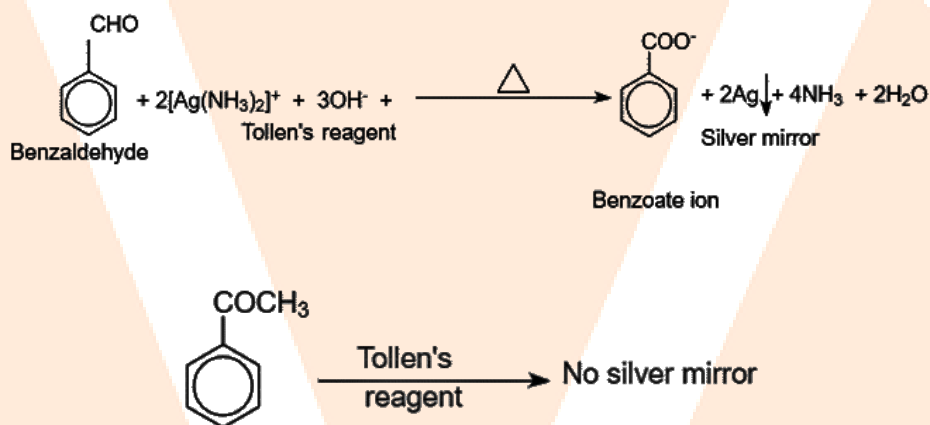
A - (Acetic Acid)

B - (Acetyl Chloride)

(b) Distinguish between:

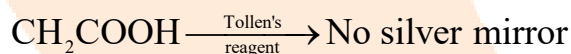
(i) $C_6H_5-COCH_3$ and C_6H_5-CHO

Ans: C_6H_5-CHO because it is an aldehyde, it reduces Tollen's reagent to produce a shining silver mirror. $C_6H_5-COCH_3$, on the other hand, does not because it is a ketone.



(ii) CH_3COOH and $HCOOH$

Ans: Ethanoic acid does not pass the silver mirror test with Tollen's reagent, whereas $HCOOH$ does.



(c) Arrange the following in the increasing order of their boiling points:

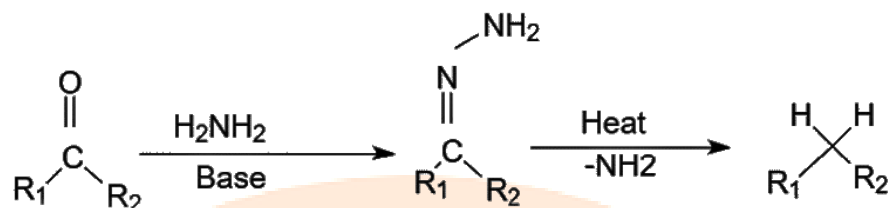
CH_3CHO , CH_3COOH , CH_3CH_2OH

Ans: $CH_3CHO < CH_3CH_2OH < CH_3COOH$

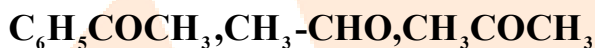
Or

(a) Write the chemical reaction involved in Wolff-Kishner reduction.

Ans: The chemical reaction is given below:



(b) Arrange the following in the increasing order of their reactivity towards nucleophilic addition reaction:



Ans: $\text{C}_6\text{H}_5\text{COCH}_3 < \text{CH}_3\text{COCH}_3 < \text{CH}_3\text{-CHO}$

(c) Why does carboxylic acid not give reactions of carbonyl group?

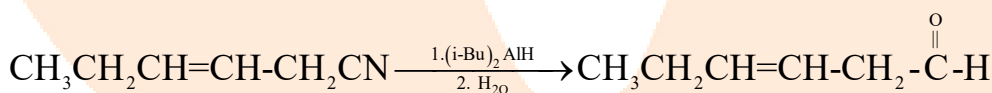
Ans: Carboxylic acid does not react with carbonyl groups because it enters into resonance with a single pair of -COOH groups, making the carbon atoms less electrophilic.



(d) Write the product in the following reaction



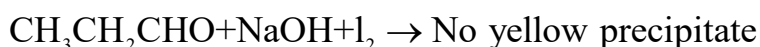
Ans: The product is:



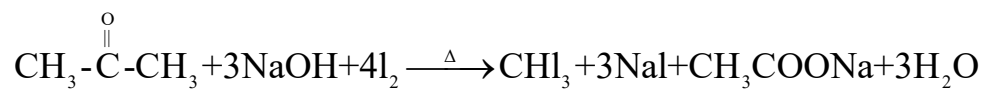
Hex-3-enenitrile Hex-3-enal

(e) A and B are two functional isomers of compound $\text{C}_3\text{H}_6\text{O}$. On heating with NaOH and I_2 , isomer forms yellow precipitate of iodoform whereas isomer A does not form any precipitate. Write the formula of A and B.

Ans:



[A] Propanal



[B] Acetone

Iodoform

(Yellow precipitate)